

function then printing an object of the Complex class wouldn't show very useful information when printed to screen.

Finally the add method takes as parameter another complex number (we just assume here, in real code it should check for this) adds it to itself and returns a new complex number that is the sum of these numbers.

Note that while variable and method names still use Snake Case in Python, class names are in CamelCase by convention.

Now let's look at class inheritance by creating a subclass of this complex class but one that adds a feature.

```
• class Complex2(Complex):
•     def __init__(self, real=0, imag=0):
•         super().__init__(real, imag)
•     def conjugate(self):
•         return Complex2(self.real, -self.imag)
```

This class derives all the functionality from the above Complex class, but adds an additional method that returns the conjugate of this complex number (same real part, negated imaginary part). Additionally it also improves the constructor by setting default values of 0 for the real and imaginary parts if no value is provided.

The new `__init__` method simply passes along the values it gets to the `__init__` method of its superclass (Complex). The original `__init__` method now handles storing the values as properties.

Virtual Environments and Python Packages

Often when working on a Python application you will need to use libraries created by other people. Python has great tools for finding and installing such packages, and then importing them into your own code to use the ready-made functionality. Django, which we will use later to build our CMS is one such package.

There can often be conflicts between different pieces of software that need different versions of the same libraries to work. To avoid this the `virtualenv` package was created, and it is now available by default with Python 3.

This package creates a virtual Python install inside a folder that once activated appears to application as the default Python install. Any packages you install while this virtual Python install is activated are installed within this folder rather than globally for the entire system.

When we develop our CMS using Django it would be wise to create a